St. Philomena's College (Autonomous), Mysore

PG Department of Biochemistry

Question Bank (Revised Curriculum 2018 onwards)

First Year- Second Semester (2019-20 Batch)

Course Title (Paper Title): Molecular Biology. QP Code:54102

UNIT	SI. NO	Questions	Marks
1	1	Highlight the contribution of Rosalind Franklin to field of Molecular Biology.	2
1	2	Differentiate between nucleotides and nucleosides.	2
1	3	When DNA molecule is subjected to agarose gel electrophoresis it moves towards anode. Why?	2
1	4	Give the composition of DNA.	2
1	5	Comment on the use of isopropanol in DNA purification.	2
1	6	Isolation of DNA from plant source is more tedious than from animal source. Why?	2
1	7	What is sugar puckering?	2
1	8	Differentiate between A and Z DNA.	2
1	9	When the relative humidity is approximately 75%, which form does a DNA molecule assume?	2
1	10	What is Hoogsteen base pair? Give its significance.	2
1	11	What is Chargaff's rule? Give its significance.	2
1	12	What are the outcomes of Hershey and Chase experiment?	2
1	13	State the central dogma of Molecular Biology.	2
1	14	What is a transforming principle?	2
1	15	Name any two viruses containing RNA as genetic material.	2
2	16	What are catenanes?	2
2	17	What are Okazaki fragments?	2
2	18	Define the role Sec A protein in DNA replication.	2
2	19	How hemimethylation prevents initiation of DNA replication at same origin?	2
2	20	What is dispersive replication?	2
2	21	Differentiate between conservative and semi conservative mode of replication.	2
2	22	What is Klenow fragment? Mention its significance.	2
2	23	What are topoisomerases?	2
2	24	Define linking number.	2

2	25	Differentiate between negative supercoiling and positive supercoiling.	2
2	26	What are DNA gyrases? Comment on its role in DNA replication.	2
2	27	What is a clamp loader?	2
2	28	Highlight the importance of magnesium ion in DNA replication.	2
2	29	Give the importance of topoisomerase inhibitors in treatment of cancer.	2
2	30	What is a replication fork?	2
2	31	What is translesion DNA synthesis? Give its significance	2
2	32	What is 'D' loop? Mention its significance.	2
3	33	Are genes and proteins collinear? Comment.	2
3	34	What is cordycepin? Comment on its role in transcription.	2
3	35	Give the role of Sigma 70 in transcription.	2
3	36	What is RNAP II holoenzyme?	2
3	37	What is a transcription bubble?	2
3	38	Differentiate between open and closed binary complex.	2
3	39	What are promoter sequences? Give an example.	2
3	40	What are upstream elements? Comment on its role in transcription.	2
3	41	What are enhancers of transcription? Comment on its role in transcription.	2
3	42	What is Pribnow box?	2
3	43	Differentiate between coding and non coding DNA strand.	2
3	44	What is a ternary complex?	2
3	45	Generally termination of eukaryotic transcription is not well defined. Why?	2
3	46	What is abortive cycling?	2
3	47	What is promoter clearance?	2
3	48	How does chain elongation takes place during transcription?	2
3	49	What is Rho factor?	2
3	50	What are sn RNA and sno RNA?	2
3	51	Highlight the importance of addition of cap and poly a A tail to m-RNA.	2
3	52	What are snRNAPs?	2
3	53	Define spliceosome.	2

3	54	What is alternative splicing? Justify alternative splicing as an agent for protein evolution.	2
3	55	What is cis and trans splicing?	2
3	56	Differentiate between RNA editing and alternative splicing.	2
3	57	Defferentiate between endonucleases and exonucleases.	2
3	58	What is RNA editing?	2
3	59	Define intron splicing.	2
4	60	What is translation? Give its significance.	2
4	61	What is genetic code? Give its importance.	2
4	62	What is the biological significance of triplet codon?	2
4	63	Which are the different features Of genetic code contributing to its versatility?	2
4	64	Assignment of codons is important for uniqueness of a genome. Justify.	2
4	65	Give the significance of studies of Khorana	2
4	66	Give the significance of studies of Nirenberg	2
4	67	What is Wobble hypothesis? Give its significance	2
4	68	Give the significance of codon Usage.	2
4	69	Which are the important Variations that occurs In The codon usage	2
4	70	What are ribosomes? Which are the different types?What is its significance?	2
4	71	What are Svedberg units?	2
4	72	Give the 3D structure Of prokaryotic ribosomes.	2
4	73	Give the 3D structure Of erokaryotic ribosomes.	2
4	74	What is the significance of ribosomal protein synthesis	2
4	75	Which are the different types of prokaryotic initiation factors? Give their significance.	2
4	76	What is shine dalgarno interaction? What is its significance?	2
4	77	What is formylation of t-RNA? Give its significance.	2
4	78	Comment on the recognition of initiator codon in prokaryotic initiation of translation.	2
4	79	What is 30s initiation complex in prokaryotic initiation? How is it formed?	2
4	80	What is 70s initiation complex in prokaryotic initiation? How is it formed?	2
4	81	What is transpeptidation reaction?	2
4	82	Which are the factors involved in translation elongation in prokaryotes? Give their significance.	2

4	83	What is the significance of proof reading in prokaryotic translation process? How is it done?	2
4	84	What is the significance of antibiotics in prokaryotic translation? Explain with one example.	2
4	85	Which are the factors involved in prokaryotic translation termination? Give their significance.	2
4	86	How does diptheria toxin work?	2
4	87	What are the factors involved in eukaryotic translation initiation? Give their significance	2
4	88	What is the significance of methylation of t-RNA in eukaryotic translation initiation? Give its significance	2
4	89	What is 48s initiation complex in eukaryotic initiation? How is it formed?	2
4	90	What is 80s initiation complex in eukaryotic initiation? How is it formed?	2
4	91	Which are the factors involved in translation elongation in eukaryotes? Give their significance.	2
4	92	Which are the factors involved in eukaryotic translation termination? Give their significance.	2
4	93	What is post termination complex of eukaryotic translation?	2
4	94	What are polysomes? Give their significance.	2
4	95	What is ribosome recycling? Give its significance.	2
4	96	What is the importance Of mRNA?	2
4	97	What are the important roles played by tRNA?	2
4	98	What is amino acyl tRNA? How is it synthesised?	2
4	99	What is mischarging of t-RNA?	2
4	100	What is the contribution of amino acyl tRNA in translation accuracy?	2
4	101	What are internal ribosome entry sites? What is its significance?	2
4	102	What is post translational modification? Give its significance.	2
5	103	What are the different types of post translational modification?	2
5	104	What is proteolysis? Give its significance.	2
5	105	What is phosphorylation? Give its significance.	2
5	106	What is glycosylation? Give its significance.	2
5	107	What is sulfation? Give its significance.	2
5	108	What is methylation? Give its significance.	2
5	109	What is hydroxylation? Give its significance.	2

5	110	What is sumoylation? Give its significance.	2
5	111	What is lipidylation? Give its significance.	2
5	112	What is acetylatyion? Give its significance.	2
5	113	What is prenylation? Give its significance.	2
5	114	What is ubiquitination? Give its significance.	2
5	115	What is disulphide bond formation? Give its significance.	2
5	116	What is signal cleavage? What is it significance?	2
5	117	What are the contributions of disulphide bond formation being a PTM? Which are the enzymes involved?	2
5	118	What is O-glycosylation? What is its significance?	2
5	119	What is N-glycosylation? What is its significance?	2
5	120	What is folding of nascent protein? What is its significance?	2
5	121	What are molecular chaperones? What role does it play in PTM of proteins?	2
5	122	Attachment of glycosyl anchor is one of the important PTM. Justify.	2
5	123	What are DNA binding protein motifs? Give examples.	2
5	124	What are zinc finger? Give their significance.	2
5	125	What are leucine zippers? Give their significance.	2
5	126	What are H-T-H motifs? Give their significance.	2
1	1	Add limelight on historical prospective of molecular biology.	5
1	2	Give the characteristics features of different forms of DNA.	5
1	3	Describe isolation and purification of DNA from plant source.	5
1	4	Explain the steps in isolation and purification of DNA from animal source.	5
1	5	How do you isolate and purify mRNA from animal source? Explain.	5
1	6	Describe Griffith’s experiment.	5
1	7	Discuss the evidences that let to conclude DNA is the genetic material.	5
1	8	Explain Avery, MacLeod, & McCarty’s experiment. What conclusion did they draw from the outcome of the experiment?	5
1	9	Discuss the outcome of the experiments that lead to conclude RNA as the genetic material.	5

1	10	Explain Frankel and Conrat experiment.	5
1	11	Give the classification of viruses based on the genetic material.	5
2	12	Write a short note on nearest neighbour base frequency analysis.	5
2	13	Explain DNA replication in E coli.	5
2	14	Explain in detail Crain's experiment.	5
2	15	Write a note on types of topisomerases and its mechanism of action.	5
2	16	Write a short note on DNA polymerase 1.	5
2	17	Give an account on structure and function of DNA polymerase III.	5
2	18	Give an account on the structure and functions of *E. coli* DNA dependent RNA polymerase.	5
2	19	Describe in detail initiation of DNA replication in *E. coli*	5
2	20	Explain termination of DNA synthesis in prokaryotes.	5
2	21	Explain the mechanism underlying in preventing re-initiation of DNA replication at the origin replication.	5
2	22	Describe in detail the initiation of eukaryotic DNA replication.	5
2	23	Write a note on fidelity of DNA replication.	5
2	24	Write a note on Trombone model of replication.	5
2	25	What is semi conservative mode of DNA replication? How was it experimentally proved?	5
2	26	What is semi discontinuous DNA replication? How was it experimentally proved?	5
2	27	Explain in detail replication of M13 bacteriophage.	5
2	28	Explain in detail replication of Lambda phage.	5
2	29	How does mitochondrial DNA replicate?	5
2	30	Give an account on rolling circle model of DNA replication.	5
2	31	Write the mechanism of action of ligase.	5
3	32	Comment on the role of CTD of RNAP II.	5
3	33	Outline the events of initiation of prokaryotic transcription.	5
3	34	Describe initiation of eukaryotic transcription.	5
3	35	Describe prokaryotic transcriptional termination.	5
3	36	Discuss rRNA biosynthesis.	5
3	37	Describe biosynthesis of t-RNA.	5
3	38	Write a short note on RNA replication of bacteriophage QB.	5

3	39	Describe prokaryotic r-RNA processing.	5
3	40	Discuss in detail t-RNA processing.	5
3	41	Discuss cap addition and poly a tail addition.	5
3	42	Detail the steps of intron splicing.	5
3	43	Describe splicing of group 3 introns and eukaryotic m-RNA primary transcript.	5
3	44	Describe processing of eukaryotic r-RNA.	5
3	45	Describe mechanism of RNA editing with examples.	5
4	46	Briefly explain the features of genetic code with example.	5
4	47	Discuss the work done by Khorana and Nirenberg.	5
4	48	Give a detailed account on evolution of genetic code.	5
4	49	Comment on the variations in usage of codons.	5
4	50	Give a detailed account on prokaryotic and eukaryotic ribosomes.	5
4	51	Briefly explain the prokaryotic translation process.	5
4	52	Briefly explain the events of prokaryotic translation initiation and the factors involved in it.	5
4	53	Briefly explain the events of prokaryotic translation elongation and the factors involved in it.	5
4	54	Briefly explain the events of prokaryotic translation termination and the factors involved in it.	5
4	55	Briefly explain the eukaryotic translation process.	5
4	56	Briefly explain the events of eukaryotic translation initiation and the factors involved in it	5
4	57	Briefly explain the events of eukaryoticc translation elongation and the factors involved in it.	5
4	58	Briefly explain the events of eukaryotic translation termination and the factors involved in it.	5
4	59	Comment on the synthesis of amino acyl tRNA and its accuracy with experimental proof.	5
5	60	Explain the different types of PTM's and their significance.	5
5	61	Discuss the events of signal cleavage. Give its significance.	5
5	62	How is disulphide bridges formed on nascent protein moieties? Give its significance.	5
5	63	Write the events of O-glycosylation on nascent proteins. Give its significance	5
5	64	Write the events of N-glycosylation on nascent proteins. Give its significance.	5

5	65	Write a detailed account on folding of nascent proteins. Give its significance.	5
5	66	What are molecular chaperones? What are the different types? Comment on its structure and significance.	5
5	67	Give the mechanism of protein folding by Chaperones. Give its significance	5
5	68	What is glycosyl anchorage? Give its significance.	5
5	69	Comment on zinc fingers and their significance.	5
5	70	Comment on leucine zippers and their significance.	5
5	71	Comment on H-T-H motif and its significance.	5

****

Question Paper Pattern- Model Question Paper

Post Graduate
St. Philomena's College (Autonomous)
MYSURU-570 015

Q.P Code: 54102

(2)

St. Philomena's College (Autonomous) Mysore
II Semester M.Sc. - Final Examination : May - 2019
Subject: BIOCHEMISTRY
Title: MOLECULAR BIOLOGY (HC)

Time: 3 Hours **Max Marks: 70**

PART –A

10×2=20

Answer any TEN of the following:

1. a. Write the composition of RNA.
   b. What are Okazaki fragments?
   c. Define triplet binding technique.
   d. What is Chargaff's rule? What is its significance?
   e. Define Wobble hypothesis.
   f. What are signal peptides?
   g. What is promoter clearance?
   h. Write the mechanism of action of ligase.
   i. What is klenow fragment? Mention its significance.
   j. Name any two viruses containing RNA as genetic material.
   k. What is 'D' loop? Mention its significance.
   l. What is RNA editing?

PART –B

Answer any FIVE of the following: **5×10=50**

2. a. How do you isolate and purify mRNA from animal source? Explain.
   b. Write an account on Hershy and Chase experiment. **5+5**
3. a. Give a note on fidelity of replication.
   b. Describe initiation of prokaryotic transcription. **5+5**
4. a. Discuss the termination of replication in prokaryotes.
   b. How is the prokaryotic transcription terminated? **5+5**
5. a. Write in detail about the processing of mRNA transcripts.
   b. Enlist the features of genetic code. **5+5**

PTO

6. a. Explain in detail about initiation in eukaryotic protein translation.

   b. Describe the role of amino acyl tRNA synthetase in translation accuracy. 5+5

7. a. How are the proteins glycosylated? Explain.

   b. Write on the role of Chaperons in protein folding. 5+5

8. **Write notes on any TWO of the following:**

   a. H-T-H motif

   b. RNA dependent DNA synthesis

   c. Trombone model of replication 5+5

****